

Project Genesis: Developer Lab Notebook

A Chronological Record of Development, Failure, and Discovery

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Abstract—This document is the developer lab notebook for Project Genesis. Unlike the Technical Architecture Reference or the Research Monograph, this document records the *process*—what was tried, what failed, what the failure revealed, what was abandoned, and what unexpected observations appeared. It is organized chronologically by development phase and is intended to serve as the project’s institutional memory. Entries written in first-person. All architectural changes documented here are cross-referenced to their formal treatment in the Research Monograph.

Index Terms—development diary, research notebook, failures, debugging, abandoned ideas, emergence, artificial life, evolutionary simulation.

I. HOW TO READ THIS NOTEBOOK

This is not documentation. It records:

- What was tried and what failed.
- What the failure revealed about the system.
- What was built as a result.
- Ideas that were abandoned, and why.
- Observations that did not fit any existing hypothesis.
- Weird bugs that revealed something true.

Entries are organized by development phase, not calendar date. Where experiment runs are referenced, their IDs match the `research/EXP-*/` directories.

II. PHASE 0 — THE BLANK GRID

A. Initial Thought

The project began with a simple observation: most simulation environments are boring. NetLogo agents live in uniform grids. They do not care about mountains, they do not get cold, they do not die beside rivers.

The question that launched everything: *What happens when you put a cognitive agent in a world that actually pushes back?*

The original sketch was not a simulation. It was a map—a real-looking map with a continent, a mountain range, a rain shadow, and rivers. And a small dot representing an agent that had to survive.

B. First Prototype

A 64×64 grid, elevation from Perlin noise, food scattered randomly, one agent moving toward the nearest food cell.

Observation: Random resource placement erases all geographic selection. If food is distributed uniformly, terrain imposes no pressure. Geography only matters if it structures resource availability.

Decision: Move to physically realistic resource distribution. Food must follow from water, which follows from rainfall, which follows from terrain. Every layer must cause the next.

III. PHASE 1 — BUILDING THE PHYSICAL WORLD

A. The Rain Shadow Bug

Early wind moisture advection produced uniform rainfall with no rain shadow. The bug: slope was computed from the full terrain height including ocean floor, so moisture climbed ocean trenches and never piled up against mountains.

Fix: Clamp terrain height to sea level (0.30) before computing wind slope. This is now the *Sea Level Wind Clamping* rule.

Lesson: Physical simulation bugs often hide in domain knowledge, not code. I needed to think like a meteorologist.

B. Priority-Flood — The Hydrology Revelation

Before Priority-Flood, rivers terminated in inland sinks. Agents could see blue on the map but could not path to those cells from certain directions because the water did not connect to the coast.

After Priority-Flood, rivers appeared that looked genuinely real—bending around hills, merging at confluences, reaching the sea.

Observation: This was the first moment the simulation felt like a *world* rather than a noise map.

IV. PHASE 2 — THE FIRST AGENTS (v1–v2)

A. Random Walkers Drown

The first agents walked randomly. Within 20 ticks, most had walked into the ocean.

Fix: Ocean cells set to infinite `movement_cost`. This introduced the movement cost grid—the first layer of environmental structure to directly constrain agent behavior.

B. Utility AI — The Over-Optimizer

When the cognitive planner was added (v2), agents walked straight to the nearest food, ignoring water even while thirsty. The planner was solving the wrong problem: nearest food, not balanced multi-need satisfaction.

Fix: Multi-objective utility vector. Each action scores against all current needs simultaneously. Highest-scoring action wins.

V. PHASE 3 — GENETICS (V3)

A. The Drift Bottleneck

Introducing genetics produced immediate rapid genetic collapse. Within 200 ticks, the population was genetically identical. The Shannon entropy of gene variance across the population dropped to ≈ 0.02 within 150 ticks.

Cause: Without spatial dispersal, agents with identical genetic starting points converged on the same resource and the first reproducer's genome dominated within three generations.

Fix: Spatial memory (Phase 4) and boredom drive (Phase 5) introduced emergent dispersal.

B. Abandoned: Dominant/Recessive Alleles

A full diploid genome with dominant/recessive alleles was implemented and tested.

Why abandoned:

- 1) Behavior was statistically identical to the haploid model at $N < 64$.
- 2) Doubled genome storage and processing cost.
- 3) Added complexity without measurable behavioral difference.

Lesson: Biological accuracy is not always the right goal. Model complexity must be justified by measurable behavioral difference.

VI. PHASE 4 — SPATIAL MEMORY (V4)

A. The Memory Clutter Bug

First implementation stored every visited cell. Within 500 ticks, each agent held 3 000–5 000 memory entries. Performance dropped from 30 to 2 ticks/sec.

Fix:

- Rolling FIFO cap: maximum 200 episodic entries.
- Spatial knowledge uses confidence-decay dict; entries below 0.0 are pruned.

Observation: Real brains have forgetting for a reason. It is computationally essential, not a design flaw.

B. Memory Confidence Decay — Getting the Rate Right

The first decay rate was too fast (forgotten within 20 ticks); the second was too slow (stale depleted coordinates retained). The final formula,

$$\Delta C = 0.0001 + g_{\text{mem}} \times 0.0009 \quad (1)$$

was arrived at empirically by observing re-exploration frequency across runs. High-fidelity agents retain spatial knowledge for $\sim 1\,000$ ticks; low-fidelity agents forget within ~ 100 ticks.

VII. PHASE 5 — EMOTIONAL DYNAMICS (V5)

A. The Oscillation Bug — Most Alarming Bug in the Project

When emotional drives were first connected to the utility engine without smoothing, agents oscillated between states every tick. An agent walking toward food would spike fear as another agent moved within 10 cells, reverse direction, drop fear below threshold, reverse again, and repeat indefinitely.

The behavioral result: agents spinning in circles at 2-tick period oscillations.

Fix: EMA smoothing ($\alpha = 0.3$) on all emotional drives. No drive can change by more than 30% of its previous value per tick.

Scientific observation: This bug was not a mistake—it revealed something real. Emotions without inertia produce pathological behavior. Real nervous systems have habituation and smoothing for exactly this reason.

B. Grief — The Emergent Emotion

Grief was not planned. While watching a run, after a mate died, the surviving agent stopped moving for 20–30 ticks with slowly declining energy. Examining drive data: `longing` was elevated from the lost social bond; `boredom` was suppressed. The interaction of drives, with no scripted grief behavior, produced something that looked like mourning.

This was left in the system as-is.

This was the first emergent behavior that surprised me.

VIII. PHASE 6 — ADAPTIVE MOTIVATIONS (V6)

A. Why Welford?

Motivation drift needs a running mean and variance of each agent's drive history without a stored history buffer. Welford's algorithm [1] computes both in $O(1)$ time and memory per update. It was implemented directly from the original 1962 paper.

B. The Motivational Deadlock

When safety and dominance motivations were simultaneously high, agents froze—oscillating between fight and flight at nearly equal utility.

Fix: Lateral inhibition. Slightly stronger drive suppresses weaker drives below threshold. Agent commits to one action.

IX. PHASE 7 — THE NEURAL PREDICTOR (V7)

A. Exploding Gradients

Continuous backpropagation after every action produced weight values of $\sim 10^8$ within 200 ticks. The predictor output became NaN.

Diagnosis: Online learning with a non-stationary distribution and no learning rate decay or gradient clipping causes gradient explosion.

Fix (Phase 7.5): Sleep-based weight consolidation. Weights update from a batch of the 20 most recent outcomes only when the agent enters rest state.

B. Does the Predictor Actually Learn?

Honest assessment at Phase 8.4: unclear.

The ablation study (EXP-XY81 vs. EXP-3N39) showed no significant survival difference with the predictor active vs. zeroed. The predictor reports 100% prediction accuracy after a few hundred ticks—which is suspicious. A $20 \rightarrow 3 \rightarrow 1$ network likely has insufficient capacity to generalize; it may

be memorizing recent outcomes rather than learning true predictive representations.

This is an open empirical question scheduled for Phase 9 investigation.

X. PHASE 8 — REFLEX LAYER, COMBAT, AND CURRENT STATE

A. The Dehydration Audit

The dehydration-beside-water problem was found not by a hypothesis, but by a visualizer observation: an agent standing in a river, thirst bar red, not moving, dying. The key telemetry entry that revealed the bug:

```
Tick 847: Agent 3 -- thirst=91.4,
action=CONTINUE_PATHING, queue_len=14
...
Tick 861: Agent 3 -- DIED: dehydration.
location=(31,47), water_cell=TRUE
```

The agent was on a water cell with 14 ticks of committed movement queued. The planner never interrupted.

Fix: Innate Reflex Layer—checks thirst tension every tick *before* the cognitive planner runs.

Observation: The most important bugs in agent systems are not crashes. They are cases where the agent does something wrong that looks almost right.

B. The Canteen Fix — An Unexpected Second Layer

After the reflex layer was introduced, dehydration did not disappear. A second audit revealed agents dying with `stored_water = 0` even after recent river visits. They drank and immediately left without filling their canteens.

Fix: The DRINK action atomically fills the canteen to maximum as a physiological byproduct. No separate cognitive decision required.

Lesson: Behavioral tuning (adjusting utility weights) cannot solve structural design problems. Some failures require architectural changes.

C. Combat Design — Five Iterations

TABLE I
COMBAT SYSTEM ITERATION HISTORY

Iter.	What changed and why
1	Binary fight/no-fight on aggression threshold. Result: 40% population loss in 100 ticks.
2	Confidence comparison added. Fights less frequent but still cascading.
3	Non-damaging Threat Display added. Most disputes resolved without injury; but winners re-challenged immediately.
4	Fear cooldown for losers. Dispute frequency dropped 78%. No extinction cascades in 10 runs.
5	Cooldown formula tuned by injury level and g_{risk} gene. Natural individual variation achieved.

D. The Cognitive Ablation Paradox

The ablation study result was genuinely unexpected. Full cognitive agents were predicted to survive significantly longer. The result: Cohen’s $d \approx 0.15$ (negligible), $p \approx 0.85$.

Initial reaction: *the cognitive system is broken*.

But spatial analysis showed completely different population structures. Full cognitive agents formed tight shelter clusters near rivers. Ablated agents died randomly across the map. The primary cause of death shifted from metabolic starvation (ablated) to territorial injuries and old age (full cognitive).

The Paradox: In a closed ecosystem with a hard population ceiling, the physical environment determines *how many* agents survive. Cognition determines *who* survives, *where* they settle, and *how* they die—not whether they die.

This forced a complete rethink of the experimental methodology. Lifespan and survivor count are insufficient metrics. Future experiments require: shelter dispersion, path entropy, trust centralization, and cause-of-death distribution as primary variables.

XI. ABANDONED IDEAS

Diploid genetics

Implemented; behavioral impact unmeasurable at $N < 64$. Abandoned to reduce complexity.

Pheromone communication

Agents emit markers at resource locations; others detect them. Abandoned because it duplicated spatial memory functionality at $O(N \times \text{area})$ computational cost per tick.

Assigned social roles (leader/follower)

Trust-centrality classifier assigned behavioral roles at spawn. Abandoned because role assignment created artificial bifurcation that prevented organic emergent leadership.

Discrete weather events

Partially implemented (temperature spikes, rainfall reductions). *Not abandoned—deferred*. System works; controlled experiments scheduled for Phase 9.

Observation: Every time a behavior was classified top-down, it broke emergence. Emergence only works if you resist the temptation to label it.

XII. WEIRD BUGS

A. Agents Celebrating Starvation

High-boredom agents selected EXPLORE over EAT with hunger at 95%. They walked purposefully away from food while dying of starvation.

Cause: Drive modulation multiplied boredom weight onto exploration utility with no cap.

Fix: Total drive modulation multiplier capped at $3.0\times$.

B. The Immortal Elder

Agent 14 survived to tick 12 000 (expected max $\sim 3\,600$ for a normal elder). Root cause: floating-point underflow in the

senescence calculation when `max_age` was extreme (due to a high `glongevity` gene), causing S to never exceed 0.001.

Fix: Hard clip ensures S always reaches 1.0 at $1.3 \times t_{\text{max}}$.

C. The UTF-8 Performance Collapse

Simulation dropped from 18 to 1.2 ticks/sec on Windows. Profiler showed the bottleneck was `stdout`, not computation.

Cause: Default Windows Cp1252 encoding on Unicode telemetry events.

Fix: Force `sys.stdout` to UTF-8; decouple telemetry to an async write queue.

Observation: Performance bugs in simulation systems are often I/O, not computation.

D. Death Density Overflow

After 5 000-tick runs, zones with many agent deaths accumulated extreme soil fertility, attracting more agents, who died, creating a self-reinforcing *death-food spiral*.

Fix: Cap `death_density_map` at 1.0; decay by 0.001 per tick.

XIII. OBSERVATIONS WITHOUT HYPOTHESES

These observations do not yet have a formal scientific framing:

- 1) **Colony color preference.** Agents from the same founding colony cluster spatially without explicit kin preference—purely because they share `home_location` and resource memory.
- 2) **Grief mimicry.** After a mate dies, agents show reduced mobility for 20–30 ticks due to elevated `longing`. Never scripted. Whether this is “genuinely” grief or a drive artifact is unanswerable in the current system.
- 3) **Territory echo.** When a colony dies out, its former territory stays unoccupied for several hundred ticks. Neighboring agents seem to avoid the space due to residual fear memories from old dispute encounters.
- 4) **The 600-tick cliff.** Most long runs show a sharp 40–60% population drop at ticks 500–700 from converging pressures: first-generation elder senescence, resource depletion from initial density, and peak territorial friction. Not a programmed event.

XIV. LESSONS LEARNED (PHASE 0 → 8.4)

- 1) **Physical models do more selection work than genetic operators.** The landscape shapes which genes survive more than mutation rates do.
- 2) **Emergence cannot be scripted.** Every time a behavior was directly programmed, it prevented something more interesting from appearing.
- 3) **Biology before cognition is not a design principle—it is a fact.** Agents without a physiological foundation died of things the brain could not help with.
- 4) **The best bugs reveal true things.** The dehydration-beside-water bug was the most important discovery in the project. It revealed a real neurological principle.

5) **Metrics must match questions.** Survival count is not the right metric for evaluating cognitive architecture. The question determines the measurement.

6) **Patience.** Most experiments take 30–60 minutes. Interrupting runs before completion wastes scientific signal.

XV. PLANNED NEXT ENTRIES

- First predator species run—design decisions and initial failure log.
- 10-seed replication of the cognitive ablation study.
- Climate volatility experiment: high vs. low seasonal amplitude.
- Cultural transmission prototype: teaching learned coordinates to offspring.

REFERENCES

- [1] B. P. Welford, “Note on a method for calculating corrected sums of squares and products,” *Technometrics*, vol. 4, no. 3, pp. 419–420, 1962.
- [2] A. R. Damasio, *Descartes’ Error: Emotion, Reason, and the Human Brain*. Putnam, 1994.